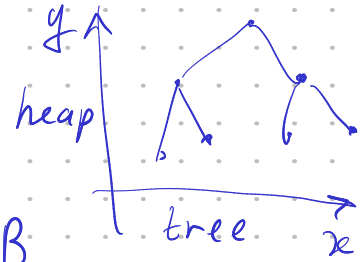
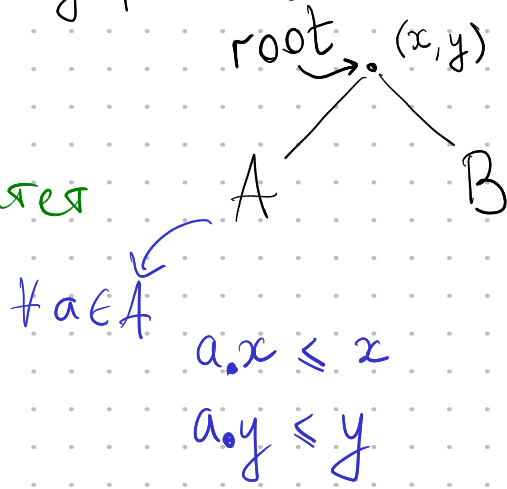


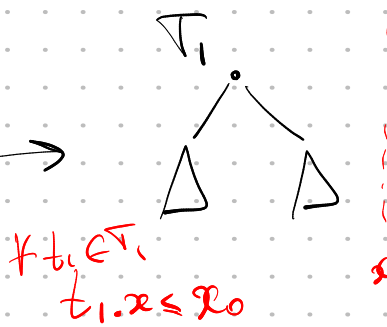
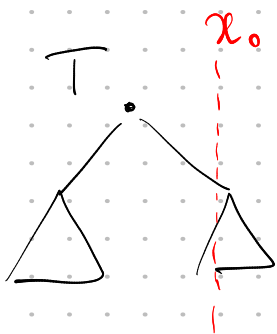
Декартово дерево (aka treap = tree + heap)

x - ключ

y - приоритет



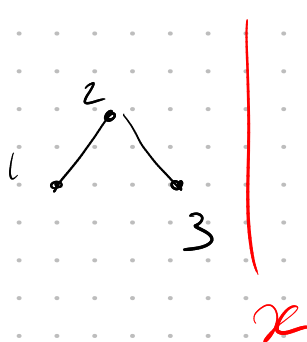
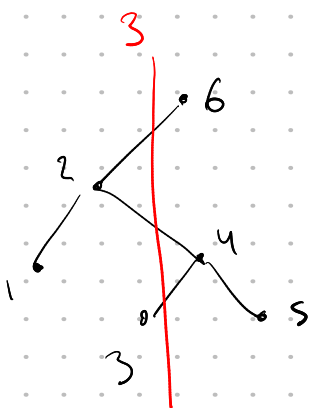
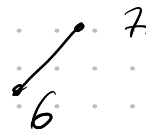
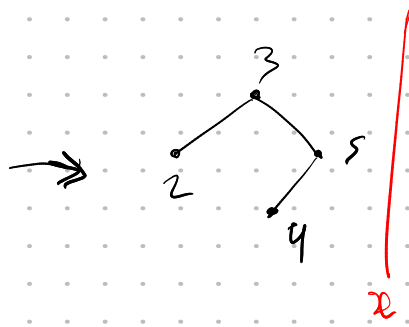
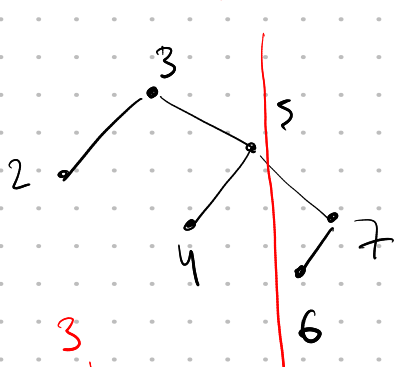
$\text{Split}(T, x_0) \rightarrow T_1 \quad T_2$



$O(h)$

$\forall t_2 \in T_2$

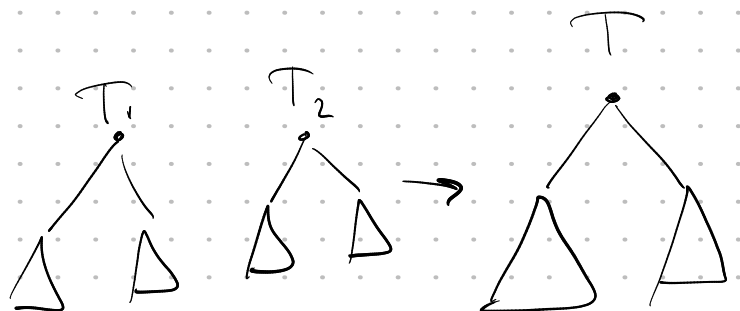
$t_2.x > x_0$



$\text{Merge}(T_1, T_2) \rightarrow T$

$\forall t_1 \in T_1; t_2 \in T_2 \quad t_1.x < t_2.x$

$O(h)$



Split(T, x_0)

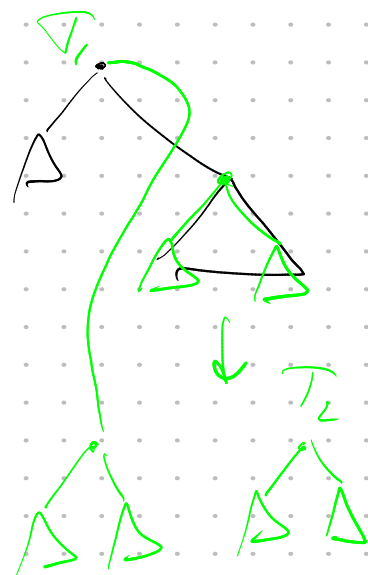
$x_0 < x$

x, y

$x_0 \geq x$

$x_0 < x$

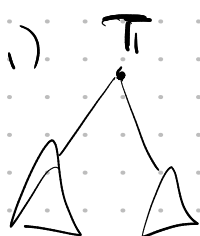
T_2



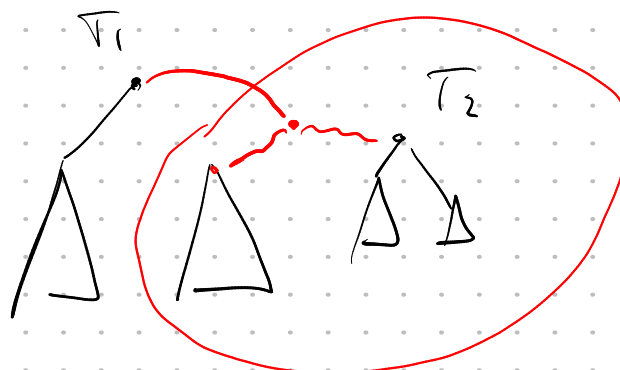
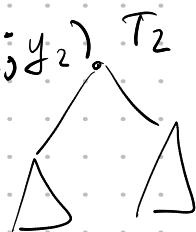
Merge(T_1, T_2)

$y_1 > y_2$

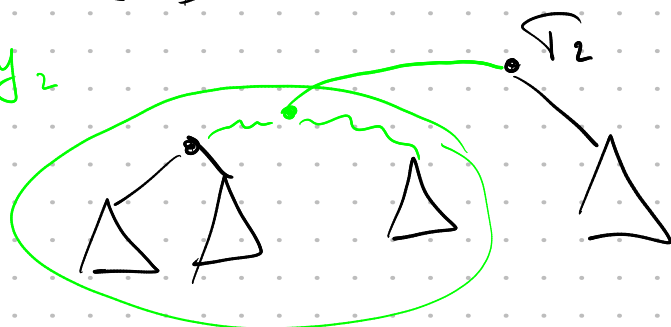
(x_1, y_1)



(x_2, y_2)

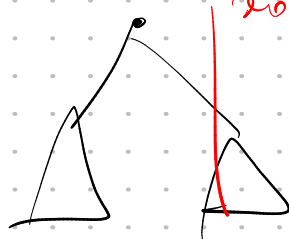


$y_1 \leq y_2$

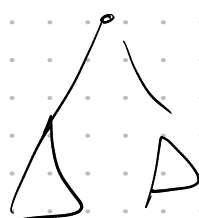


$+x_0$

x_0



l



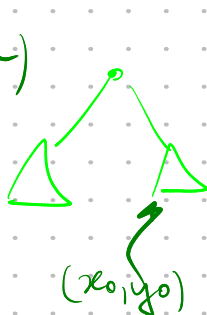
x_0, y_0

m

r



merge(merge(l, m), r)



Если $y \sim R$ (случайные), $h \approx \log_2 n$

`set<int>  $O(\log_2 n)$`

`treap<int>  $y = \text{rand}$   $O(\log_2 n)$`

DP по явному ключу y - тот же самый
 x - кол-во вершин в поддереве

